

# AI Use Log

## Scope and authorship

This log records AI-assisted preparation of the Matco Foods / Falak Extreme 5 kg assignment. The student selected the company and product, accepted documented synthetic assumptions, required the PDF’s synthesis prompt, and requested completion into seven files. AI generated the dataset, code, automated checks, charts, dashboard and draft narrative. The student’s own manual checking and understanding are not claimed here.

| Stage / date             | AI-generated work                                                                                  | Checks or corrections                                                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 30 Sep 2026: synthesis   | 156 weekly observations; dictionary; assumption log; Section 4.2 prompt completed for Matco.       | Regular shelf price vs promo depth defined. Revenue benchmark distinguished from assumed SKU sales. Dataset frozen; seed/hash recorded.                              |
| 01 Oct 2026: diagnosis   | Descriptive statistics, annotated sales chart and 135-word diagnosis.                              | Independent statistics and grouped counts checked. Promotion/event overlap separated; CV not treated as safety stock.                                                |
| 03 Oct 2026: forecasting | Seven common-origin candidates; metrics; holdout chart; five largest misses.                       | Recursive SMA uses predictions. Historical weekly planning forecasts scored separately. Model/state formulas matched notebook results.                               |
| 05 Oct 2026: completion  | Regression, forward forecast, empirical envelope, operational scenarios and seven final artifacts. | Centered/scaled QR cross-check; HAC/HC3 uncertainty; capacity and cost assumptions explicit; inventory conservation checks; metric Excel formulas; cross-file audit. |

## Tools and execution

ChatGPT/Codex assistant; Python with pandas, NumPy, SciPy and Matplotlib; JavaScript spreadsheet authoring; reportlab PDF creation; Poppler PDF rendering; self-contained HTML/SVG/JavaScript dashboard. No exact model version is asserted. No paid scraping service was used for this assignment.

## Company and product calibration

The frozen assumption log records FY2026 reported group sales of PKR 23,846.71 million and a 30 Sep 2026 Falak-store price snapshot of PKR 2,865 regular / PKR 2,800 offer. Completion rechecked the reported sales via MarketScreener and the separate PSX annual sales table. The issuer filing was located but scanned consolidated text was not independently extracted. These company-wide and retail context facts do not establish SKU sales or costs.

Sources: MarketScreener FY2026 result: <https://www.marketscreener.com/news/matco-foods-limited-reports-earnings-results-f-or-the-full-year-ended-june-30-2026-ce7858dfd08bf025>  
PSX: <https://dps.psx.com.pk/company/MFL>  
Falak product snapshot:  
<https://estore.falakrice.com/collections/basmati-rice/products/extreme-basmati-rice-5kg-free-recipe-mixes>

# Prompts and specifications used

## A. User directions recorded in the conversation

“my company is Matco Foods with product Falak Rice”  
“lets do with falak extreme 5 kg the other two are fine”  
“use the documented assumptions ... also use the prompt given in the pdf for the dataset”  
“okay now complete this assignment and give me 7 deliverables in 7 seperate files”

## B. Section 4.2 synthesis prompt, completed and retained

The following is the preserved dataset-generation specification. It was retained from the original dataset workbook, not replaced with a new prompt designed to improve forecast scores.

I am a BBA student in a Supply Chain Analytics course. I have selected Matco Foods Limited, a company listed on the Pakistan Stock Exchange, as my company for a semester project. I need you to generate a SYNTHETIC, illustrative weekly demand dataset for ONE of their products, so I can practice forecasting, regression, and validation.

PRODUCT / FORECAST OBJECT: Falak Extreme Basmati Rice, 5 kg packs, Pakistan domestic retail

REAL CONTEXT TO CALIBRATE FROM (do not invent different real facts about the company):

- Company's most recently disclosed annual revenue: approximately PKR 23.84671 billion consolidated sales for the year ended June 30, 2026 (reported August 28, 2026)

- Broad industry/sector: Food processing / branded packaged rice (FMCG)

- Company's fiscal year-end: June 30

GENERATE 104 TO 156 WEEKLY OBSERVATIONS WITH THESE COLUMNS:

- Week\_Start\_Date (consistent weekly dates)
  - Actual\_Demand (units sold or ordered — this is the forecast target)
  - Historical\_Forecast (a plausible prior forecast series, for error-metric practice)
  - Price (numeric, varying over time — not flat)
  - Promo\_Flag (0/1) and Promo\_Depth (% discount when active)
  - Event\_Flag (0/1) tagged to a calendar event appropriate to my sector — for FMCG or food, use Ramadan/Eid; for fashion, use a seasonal launch/markdown window; for cement/construction, use the spring-summer building season; for pharma, use a disease-season or regulatory-driven spike; choose the closest fit for my sector
  - One additional driver relevant to my sector: Temperature, Distribution\_Points, Marketing\_Spend, Competitor\_Activity, or Economic\_Index — pick the most plausible one and justify your choice in one sentence
  - Stockout\_Flag (0/1) for weeks where availability constrained real demand
  - Exceptional\_Event (text, blank in most weeks) — flag 2 to 4 one-off shocks such as a competitor promotion, a supply disruption, a weather event, or a viral marketing moment
- THE DATA MUST HAVE THESE REALISTIC CHARACTERISTICS, NOT RANDOM NOISE:
1. A believable base LEVEL and TREND (growing, flat, or declining) appropriate to the sector — state your reasoning in one sentence
  2. SEASONALITY appropriate to the sector, as specified above for Event\_Flag
  3. A PRICE effect: weeks with higher price should show a modest, imperfect demand reduction — not a mechanical, identical relationship every time
  4. A PROMOTION effect: promotional weeks should show a visible demand lift (10-40% above baseline), varying in size week to week — not the exact same lift every time
  5. Natural WEEK-TO-WEEK NOISE — real demand is never a smooth curve
  6. The 2-4 EXCEPTIONAL EVENTS should each visibly disrupt demand for 1-2 weeks, then the series should return to its normal pattern

7. Ensure NO relationship is deterministic. A promotion must not increase demand by exactly the same amount every single time it runs.

AFTER THE TABLE, WRITE:

- 1. A short "Assumptions" paragraph (5-7 sentences) explaining the starting demand level, the trend, the seasonal pattern, the price and promotion effect sizes you built in, the additional driver you chose and why, and what this dataset does NOT capture.
- 2. A Data Dictionary table: Field | Definition | Unit | Real / Assumed / Synthetic | Why Included.

OUTPUT the data as a table I can copy directly into Excel or a CSV file.

MANDATORY LABELING — include this exact sentence at the very top of your response, before anything else:

"Company-Calibrated Synthetic Dataset for Academic Analysis — illustrative figures, calibrated to Matco Foods Limited's disclosed scale, not actual company records."

ADDITIONAL PRODUCT-SPECIFIC INSTRUCTIONS AND AGREED ASSUMPTIONS

Generate exactly 156 Monday-start weeks from 2023-10-02 to 2026-09-21. One unit is one 5 kg pack. Use 5,000 baseline weekly packs as an assumption, justified by 1,000 assumed outlets selling five packs each.

Choose Distribution\_Points as the additional driver: wider retail availability can support higher total pack sales. Include Product\_or\_Category, Market\_Channel, Event\_Name and Data\_Split so product, scope, calendar labels and the validation split are explicit.

Use the documented Pakistan Ramadan/Eid dates and aggregate daily shopping windows to weeks. Use short stocking/Eid peaks and a smaller middle-Ramadan lift. Do not use an arbitrary fixed week each year.

Define Price as regular shelf price before promotional discount. The historical price path is assumed, anchored to the current official regular price of PKR 2,865 and offer price of PKR 2,800. Store Promo\_Depth as a fraction: 0.10 = 10%.

Use a modest 2% annual underlying trend, stepwise outlet expansion, 36 promotional weeks with 5-12% discounts, imperfect 10-40% promotion lifts and about 7.5% ordinary noise. Allow observed net effects to differ because other drivers and stockouts can mask uplift.

Create four labelled two-week illustrative shocks: competitor discounts, retailer displays, packaging delay and transport disruption. Mark supply-constrained weeks with Stockout\_Flag. Never present these as actual Matco events.

Historical\_Forecast must be a plausible prior planning series that uses no current or future Actual\_Demand. Label the first 144 weeks Train and final 12 Holdout; do not fit forecasting models yet.

Attach the mandatory disclosure as Dataset\_Label to every CSV record. Freeze the output before model comparison. Record the random seed and a CSV hash. The company revenue is a scale benchmark only, and it cannot establish actual SKU demand or company net revenue from retail value.

## C. Forecasting specification from Section 8

Use `train_df` and a 12-week `test_df`. Build naive and seasonal naive; a four-week moving average; SES at alpha 0.3 and 0.6; and a seasonal/trend-capable method accounting for `Event_Flag`. Fit only training data, do not use held-out actuals recursively, and plot all candidates against held-out actuals. The implementation uses event-adjusted damped Holt as the justified calendar-aware replacement for forced fixed-week seasonality.

## D. Regression specification from Section 10

Using `train_df`, regress `Actual_Demand` on `Price`, `Promo_Flag`, `Event_Flag` and `Distribution_Points`. Report intercept, every coefficient,  $R^2$ , adjusted  $R^2$  and p-values. Explain each association holding other included variables constant. Add uncertainty and omitted-variable caveats.

## E. Dashboard and decision specification

Create one interactive executive dashboard covering history, future demand, model comparison, errors, driver associations, scenarios and a quantified action with owner/risk/review trigger. Use the final results without inventing company capacity, costs or causal claims. Make a six-frame storyboard and a one-page memo consistent with that decision.

## F. Audit specification

Check date ordering, source hash, train/test separation, leakage, recursive logic, MAD, MAPE, RMSE, WAPE, Bias, RSFE and Tracking Signal. Recalculate key metrics independently, identify five largest misses, challenge regression interpretation, and test which operational assumptions could reject the recommendation. This was implemented through programmatic checks and a separate validation note, not a separate human or external auditor.

## Observed limitations and corrections

Holt's training fit selected zero trend rather than a forced trend. Price significance was not overstated. The event coefficient was treated as an association. A descriptive residual envelope was labeled without a confidence or service guarantee. The operating cost was separated from retail price. Delay/interruption recovery was explicitly capacity-bounded and shortages were not automatically valued as lost sales. Dashboard values and narrative were generated from the same results. A visual date-label check found a timezone shift; parsing date-only values in UTC corrected the labels before delivery.

## Student review before submission

Review the source assumptions and company references, calculate at least three metrics independently by hand or by inspecting Excel formulas, explain one regression coefficient and its caveats, demonstrate the dashboard controls, and be able to defend the decision. Use this AI log in the submission. These student actions are required by the PDF's self-check and are not marked completed on the student's behalf. The package does not make any claim of "0% AI".

## Submission route

The assignment specifies LMS and GitHub submission on 05 Oct 2026. This package provides seven files for that submission; no LMS upload, GitHub publication or message to the instructor has been performed.

Technical references: NumPy least-squares documentation ([numpy.org/doc/stable/reference/generated/numpy.linalg.lstsq.html](https://numpy.org/doc/stable/reference/generated/numpy.linalg.lstsq.html)); SciPy Student-t distribution documentation ([docs.scipy.org/doc/scipy/reference/generated/scipy.stats.t.html](https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.t.html)).